

# ADVANCED LINEAR ALGEBRA

## HOMEWORK 3

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**P.1.32** (a) Give an example of a nonzero  $A \in \mathbf{M}_2(\mathbb{F})$  such that  $\text{col}(A) = \text{null}(A)$ . (b) Give an example of  $A \in \mathbf{M}_2(\mathbb{F})$  such that  $\text{col}(A) \oplus \text{null}(A) = \mathbb{F}^2$ .

*Solution.* Sol

□

**P.1.39** In  $\mathbf{M}_n$ , let  $\mathcal{U}$  denote the set of strictly lower triangular matrices and let  $\mathcal{W}$  denote the set of strictly upper triangular matrices. (a) Show that  $\mathcal{U}$  and  $\mathcal{W}$  are subspaces of  $\mathbf{M}_n$ . (b) What is  $\mathcal{U} + \mathcal{W}$ ? Is this sum a direct sum? Why?

*Solution.* Sol

□

**P.1.41** Let  $f \in C[-1, 1]$ . Then if  $f$  is *even* if  $f(-x) = f(x)$  for all  $x \in [-1, 1]$ , and  $f$  is *odd* if  $f(-x) = -f(x)$  for all  $x \in [-1, 1]$ . Let  $\mathcal{U}$  denote the set of even functions in  $C[-1, 1]$  and let  $\mathcal{V}$  denote the set of odd functions in  $C[-1, 1]$ . Show that (a)  $\mathcal{U}$  and  $\mathcal{V}$  are subspaces of  $C[-1, 1]$ , and (b)  $C[-1, 1] = \mathcal{U} \oplus \mathcal{V}$ . *Hint.*  $f(x) = \frac{1}{2}(f(x) + f(-x)) + \frac{1}{2}(f(x) - f(-x))$ .

*Solution.* Sol

□

**P.1.44** Let  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{C}^n$  and let  $A \in \mathbf{M}_n$  be invertible. Prove that  $A\mathbf{v}_1, A\mathbf{v}_2, \dots, A\mathbf{v}_r$  are linearly independent if and only if  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{C}^n$  are linearly independent.

*Solution.* Sol

□

**P.2.1** A full-rank factorization is not unique. For example, let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

show that the following two decompositions work for  $A$

$$\begin{bmatrix} 5 & 8 \\ 23 & 38 \\ 41 & 68 \end{bmatrix} \begin{bmatrix} 1 & 6 & 11 \\ -\frac{1}{2} & -\frac{7}{2} & \frac{13}{2} \end{bmatrix}, \begin{bmatrix} 14 & -19 \\ 32 & -43 \\ 50 & -67 \end{bmatrix} \begin{bmatrix} \frac{11}{2} & \frac{3}{2} & -\frac{5}{2} \\ 4 & 1 & -2 \end{bmatrix}$$

*Solution.* Sol

□

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**P.2.2** Let  $A \in \mathbf{M}_n$  with  $a_{ij} = 1$  if  $i = j$  or  $i + j = n$ , and  $a_{ij} = 0$  otherwise. Compute  $\text{rank}(A)$ , and find a basis for  $\text{col}(A)$ .

*Solution.* Sol □

**P.2.3** Let  $A \in \mathbf{M}_n$  with  $a_{ij} = 1$  if  $i + j$  is even and  $a_{ij} = 0$  if  $i + j$  is odd. Compute  $\text{rank}(A)$  and find a basis for  $\text{col}(A)$ .

*Solution.* Sol □

**P.2.4** Let  $A \in \mathbf{M}_n$  with  $a_{ij} = 1$  if  $1 \in \{i, j\}$  or  $n \in \{i, j\}$ . Compute  $\text{rank}(A)$  and find a basis for  $\text{col}(A)$ .

*Solution.* Sol □

**P.2.5**  $\mathbf{x} \in \mathbb{F}^m$  and  $\mathbf{y} \in \mathbb{F}^n$  be nonzero vectors. Compute  $\text{rank}(\mathbf{xy}^\top)$

*Solution.* Sol □

**P.2.6** Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbf{M}$  and suppose that  $\dim(\text{row}(A)) = 1$ . If  $[a \ b] = \lambda[c \ d]$ , how are the columns  $\begin{bmatrix} a \\ c \end{bmatrix}$  and  $\begin{bmatrix} b \\ d \end{bmatrix}$  related?

*Solution.* Sol □